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CancerOS — System Architecture & Infrastructure Blueprint

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Version: 1.0
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Status: Proposed, implementation-ready architecture with explicit verification gates
Scope: Breast-cancer research and hackathon proof of concept; **not a clinical decision-support or treatment-planning system**

1. Purpose and engineering position

CancerOS is a research prototype that combines verified breast MRI data, the available structured clinical record, reproducible hypothetical longitudinal trajectories, lesion geometry, and evidence-linked AI interpretation. The required demonstration includes a real imaging-derived baseline, five separately labeled synthetic scenario families, dated clinical observations, rotatable tumor geometry, and at least one independently sourced genuine longitudinal comparison where a suitable case can be validated.

This blueprint turns the accompanying PRD and specifications into an executable technical architecture. It **does not** treat the documents' aspirations as evidence of available datasets, model accuracy, clinical validity, cloud entitlements, or production readiness. Every externally obtained artifact must have a documented identity, license, checksum, appropriate source linkage and data-quality result before use.

Architectural decision: build a **modular monolith plus asynchronous workers**, not a microservice estate. Use Python/FastAPI for backend and scientific computation, Next.js/TypeScript for the UI, PostgreSQL for typed research data, private object storage for MRI-derived artifacts, and a queue for computational jobs. Keep optional segmentation GPU inference and language-model inference behind adapters. This is simpler to test, cheaper to deploy, and sufficient for the proposed 100-baseline / 500-branch *planning* ceiling; the first demo needs only a smaller verified cohort.

1.1 Core correctness invariants

1. Authentic observations and source images are immutable after import; changes create versions or append-only corrections.
2. Genuine longitudinal visits, contrast phases in a single MRI study, synthetic branches, and model-predicted future states have distinct types.
3. No synthesized lesion mesh can be represented as an actual follow-up MRI; images and derived geometry must declare their origin.
4. An image-analysis mask is not automatically an expert anatomical lesion mask. Annotation purpose and quality class control which measurements are permitted.

5. Numeric measurements and time differences come from deterministic scientific code, **not** the language model.
6. Unknown clinical attributes remain unknown. No automated inference of stage, ER/PR/HER2 status, or treatment benefit without adequate source evidence.
7. All displayed numeric or narrative findings link to a patient state, an original or derived artifact, a model version and appropriate provenance.
8. External AI only receives an explicitly permitted, minimal, de-identified payload after policy checks; no unrestricted access to database or source scans.
9. Hypothetical treatment exposure is not evidence that treatment caused an outcome.
10. Any process that fails validation produces a visible failure state, never a plausible-looking fabricated result.

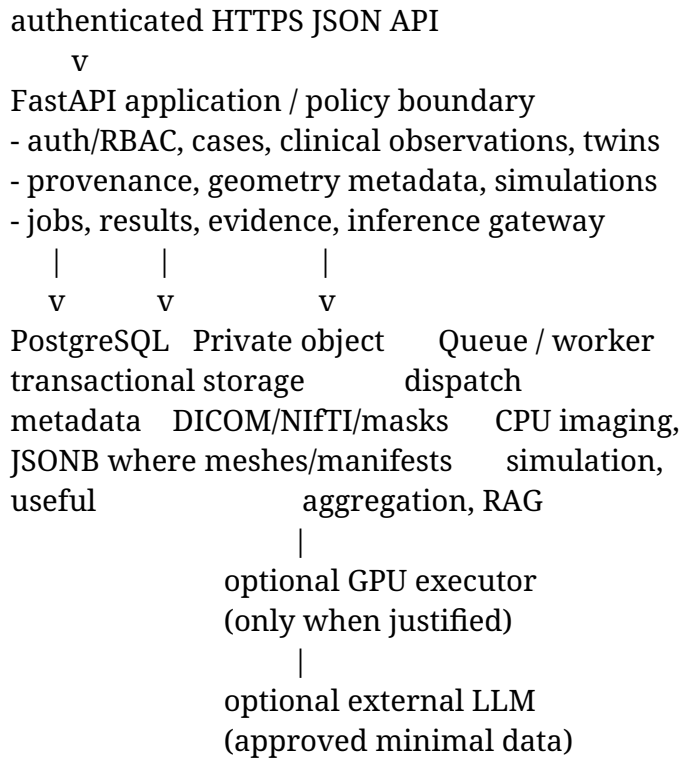
2. Verification ledger: documented, checked and unresolved

Claim	Evidence / qualification	Implementation decision
I-SPY2 contains repeated breast MRI and clinical metadata	TCIA collection records 719 core cases; 985 in broader imaging cohort and separate clinical data. Not every case has every usable visit.	Source of preferred real longitudinal demo; check patient-level image, time and spreadsheet linkage.
Duke MRI and BreastDCEDL are usable without restriction	Duke and BreastDCEDL are CC BY-NC 4.0 according to source listings.	Use only under permitted hackathon/research conditions; separately review use in public or commercial demonstrations and redistribution.
BreastDCEDL contains ready-to-use 3D data / masks	Verified at dataset-description level; suitability and anatomical meaning vary by file. Derived from underlying cohorts, **not** an independent cohort.	Use as an accelerator; preserve parent source and prohibit train/test leakage.
nnU-Net v2 or MONAI equals an available clinically validated checkpoint	These are frameworks/architectures; a matching checkpoint, input-channel contract and held-out performance are not guaranteed.	P0: import suitable verified masks; P1: model proposal after evaluation.

Claim	Evidence / qualification	Implementation decision
Meta Muse Spark through Model API	Meta publishes available model IDs, API capabilities and service documentation. Account access, current cost, policy, data retention and medically relevant reliability require checking.	Provider-agnostic LLM gateway; use configured ID, deny external PHI export by default.
Meta's image segmentation API can replace MRI tumor segmentation	Generic image/video segmentation support does not establish validation for volumetric DCE-MRI lesions.	Do not use as the default medical segmentation system.
Five scenarios describe realistic personalized clinical outcomes	Input document explicitly presents these as modeling templates; no patient-specific calibration established.	Label simulation as hypothetical, use rule-defined parameters and reproducible seeds.
GPU is mandatory	Not for mask import, mesh computation, relational data, visualization and deterministic simulation.	Local CPU-first; optional on-demand GPU only after a segmentation benchmark.
100 patients / 500 branches / 2,000 visits are already available	Planning targets only, as expressly stated in the documents.	Capacity envelope, not minimum hackathon success criterion.
Cloud services are free or a stated amount	Pricing depends on region, account credits, usage and egress/requests.	Model costs by measured units, budget alerts, no unsupported fixed-dollar guarantee.

3. System context and trust boundaries

RESEARCH USER / REVIEWER / ADMIN
|
TLS
v
Next.js frontend (browser)
- cases, 3D twin, timelines, lab trends, evidence
|



Trust boundaries: browser is untrusted; public API authenticates and authorizes every record; storage buckets are private; job workers operate with least-privilege credentials; external AI provider is a separate data-processing boundary; imported datasets are untrusted inputs and must be isolated until validation. A researcher-approved synthetic demo asset may be served with a short-lived signed URL; original research MRI or unreviewed metadata must not be public.

3.1 Top-level data flows

Import: approved source manifest → quarantine object area → checksum/license/de-identification and spatial checks → import event → canonical relational records → curated immutable source artifacts.

Imaging: validated MRI + mask → versioned preprocessing → segmentation-mask QC → physical measurements → surface extraction / mesh optimization → rendered artifacts + geometry and provenance record.

Simulation: baseline snapshot → scenario eligibility → deterministic seeded branch generation → numerics, dates and clinical-consistency checks → synthetic observations and distinct geometry → publication only if validated.

Interpretation: authorized patient-state snapshot → deterministic statistics + source references → retrieved curated evidence → minimal prompt to configured LLM → schema validation, claim/source checking and unsupported-claim rejection → versioned research interpretation.

Viewer: case selection → authorized summary, timelines and artifact manifests → short-lived asset URLs → 3D/2D client rendering; no direct model, database or secret access from the browser.

4. Frontend architecture

Framework: Next.js with TypeScript, React, Tailwind or equivalent accessible component library. Prefer server-rendered routing and non-sensitive summaries where appropriate; load heavy interactive 3D viewers client-side. Avoid moving MRI measurements to browser code.

Feature areas: (a) Case library and filter; (b) research-patient summary, diagnosis and provenance; (c) twin workspace (rotatable breast-region lesion mesh, optional context geometry, synchronized date slider); (d) observed vs. synthetic branch comparison; (e) clinical measurements/laboratory/vital/symptom/treatment graphs; (f) AI interpretation with evidence links and uncertainty; (g) admin ingest, QC, jobs and validation panels.

Visual semantics: persistent source labels OBSERVED, DERIVED, SYNTHETIC, INTERPOLATED, PREDICTED; labels must not rely solely on color. At each visit show timestamp, source type, geometry type, annotation quality and missing modalities. If no lesion mesh is supported, show the valid imaging or measurement without a fabricated 3D lesion.

Imaging rendering options: three.js / React Three Fiber for compact indexed GLB/glTF lesion geometry; Cornerstone3D for slice viewing. OHIF is mature for DICOMweb and uses Cornerstone3D, but embedding the full viewer into a hackathon dashboard increases integration complexity. Choose an OHIF workflow only if proper DICOMweb service and viewer use cases justify it.

Client constraints: progressive mesh download, level of detail, camera fit, geometry bounds, optional image slices, loading/retry states, WebGL fallback, keyboard navigation, and explicit disclosure when surrounding breast contour is illustrative rather than patient-derived. Do not expose object-bucket credentials in web code.

5. Backend modules and service boundaries

Keep one deployable FastAPI application with separate packages and repository interfaces; deploy heavy workers independently.

Module	Ownership	Explicit non-responsibility
identity_access	Auth integration, RBAC, audit attribution, patient-level authorization	Clinical decision making
source_registry	Dataset license, source patient map, import manifest, checksums	Silent clinical field completion
clinical_records	Diagnosis, visits, lab results, vitals, notes,	Simulated values mixed into originals

Module	Ownership	Explicit non-responsibility
	treatment, outcome	
imaging_catalog	Studies, acquisitions, DICOM/NIfTI, geometry QC, mask types	LLM reading image pixels as quantitative evidence
geometry	Physical-space masks, validated volumes, mesh artifacts, registrations	Claiming simulated meshes are real scans
longitudinal	Authentic state chronology, comparisons, feature aggregation	Treating within-study phases as separate visits
simulation	Scenario eligibility, run seed, branch outputs, simulation checks	Predicting actual patient outcome
evidence	Versioned literature corpus, citation/source lookup, vector retrieval	Replacing original clinical observations
ai_gateway	Approved provider adapter, request policy, structured output, citations	Arbitrary database queries and autonomous treatment advice
jobs	Job lifecycle, idempotency, retries and resource limits	Concealing failed tasks
audit_provenance	Immutable lineage events, result versions, model/config identities	Raw data editing

Use typed Pydantic request/response objects, OpenAPI definitions, SQLAlchemy/Alembic for relational persistence and explicit domain service methods. High-cost tasks return a job ID, not an open HTTP request held until MRI processing completes.

6. Database schema and state model

Primary store: PostgreSQL; use normalized tables for fundamental research records and JSONB only for clearly versioned, bounded model/simulation configurations. Use UUIDs as internal IDs, FK constraints and indexed observation timestamps. The schema below maps to the attached data specification while adding the systems concerns missing from that document.

Source domain: dataset_sources, source_patient_mappings, research_patients, source_artifacts, licenses, import_batches, processing_manifests, provenance_records.

Clinical domain: cancer_diagnoses, clinical_visits, laboratory_results, vital_signs, symptom_observations, clinical_notes, treatment_events, outcome_observations.

Imaging domain: imaging_studies, imaging_series, imaging_acquisitions, segmentations, lesions, tumor_measurements, registrations, geometry_artifacts, imaging_qc_results.

Longitudinal domain: patient_states (state_type, event_time, source_visit_id, branch_id, status), state_observation_links, simulation_branches, simulation_runs, simulation_parameters, synthetic_observations, validation_results.

AI/evidence domain: evidence_documents, evidence_passages, evidence_embeddings (optional), extracted_facts, model_results, interpretations, interpretation_evidence_links, model_registry.

Operations domain: users, role_assignments, jobs, job_attempts, audit_events, asset_grants (if needed), schema_versions.

Critical constraints:

- Enforce patient_states.state_type as a controlled enumeration. source_visit_id must correspond to a genuine visit for observed states; branch_id must exist for synthetic descendants. Cross-branch joins must check patient parent identity.
- imaging_acquisitions belong to imaging_studies; DCE contrast phases do **not** manufacture new visit records.
- tumor_measurements require method, physical unit, segmentation or imaging reference, version and QC state. Do not compare non-equivalent lesion definitions without an explicit compatibility flag.
- Synthetic branch records include parent baseline, scenario family, configuration hash, random seed and version. A run never overwrites its originating observations.
- Evidence-linked interpretation references must resolve to persisted source/derived/evidence records; unknown source IDs cause publication failure.
- Every artifact has SHA-256, mime/format, size, URI/key, license identity, generation version, parent IDs and availability status.
- Never store direct patient identifiers, names or invented dates of birth. Protect pseudonym linkage and remove patient-bearing image tags as permitted by source agreements.

Migration approach: numbered Alembic migrations, unique constraints for source identity, read-only materialized snapshots for viewer aggregation when needed, scheduled backups and restoration drills. A dedicated knowledge-graph database is not necessary for the initial source/evidence graph: relational edge tables plus indexed queries suffice.

7. Object storage, formats and lifecycle

Use private bucket(s): raw-quarantine, curated-originals, processed-volumes, masks, meshes, reports-evidence, temporary-exports (logical prefixes or separate buckets based on access policy).

Artifact	File representation	Notes
MRI original	DICOM with protected import metadata	Immutable; verify tags, geometry, legal use and checksum
Research-ready volume	NIfTI .nii.gz plus affine, processing JSON	Never drop spatial orientation or provenance
Lesion mask	NIfTI or suitable segmentation DICOM	Store mask meaning/class, label map and spatial correspondence
Web geometry	.glb/glTF with sidecar metadata	Optimized, physical scale preserved; explicitly source-derived or synthetic
Longitudinal output	PostgreSQL structured rows, optional JSON/Parquet export	Branches share a lineage to original patient
Model and QC output	Versioned JSON with schema ID	Include input hashes, software/model version and QC
Publication and literature	PDF/text/chunks with rights and metadata	Only approved source corpus and lawful excerpts

Publish using a two-stage procedure: write a versioned object and verify checksum → commit relational artifact record → mark published only once integrity/authorization checks pass. If an operation fails after object upload, cleanup or quarantine orphaned objects. Use short-lived signed reads. No public-bucket defaults, and no anonymous access to raw images.

Scale calculation: neither 100 baselines nor 500 branches determines storage without sampling file sizes. Estimate total = originals retained + processed volumes + masks + meshes + lineage + backups. A 1.75 TB source archive need not be mirrored for a five-case demo. Measure representative patient sizes before buying storage. Browser distribution should mainly deliver compact meshes, not full DICOM series.

8. MRI / segmentation / twin computation

CPU pipeline:

1. Read approved DICOM or NIfTI and associated metadata; validate source study, series, geometry and required image channels.
2. Construct a physically meaningful volume; sort DICOM slices by geometry rather than filename.

3. Normalize orientation with stored LPS/RAS transforms; retain source affine. Resample only for documented downstream requirements; interpolate masks with label-preserving interpolation.
4. Import reference lesion mask (P0) and check that dimensions, affine, lesion semantics, connectivity and registration are appropriate. A functional tumor-volume mask must remain described as a functional analysis region where applicable.
5. Calculate voxel-volume-derived lesion volume n_mask_voxels × |det(3×3 affine spatial matrix)| in mm^3, convert to cm^3 by dividing by 1,000. Derive any diameter by an explicitly documented physical-space algorithm, not just raw pixel indices.
6. Extract surface using marching cubes in physical coordinates; validate surface geometry, smoothing tolerance and coordinate handedness; export web-compatible mesh with scale metadata.
7. Associate only comparable measurements across verified dates. Store registration quality and the choice of reference space if scans are aligned.
8. Render and evaluate representative MRI/mask overlays and 3D views before accepting a patient asset.

Segmentation GPU path (P1): only when a model checkpoint is available, accepts the MRI sequence/channel scheme, has known preprocessing and is evaluated on patient-independent held-out annotations. Record Dice, IoU and surface distance such as HD95 where meaningful; do not imply a clinically acceptable threshold without independent specialist validation. Model-derived contours must be visually inspectable and independently flagged as proposals.

Synthetic geometry: apply configured geometric operations to a baseline-derived mask/mesh consistent with the stated simulated burden. Preserve topology/quality checks and avoid implying that simple geometric scaling models true tumor invasion or response. The generated product is *synthetic geometry*, not a synthetic authentic MRI scan.

9. Synthetic longitudinal engine (SLRE)

Use a deterministic, versioned Python computation package, **not** an LLM writing arbitrary follow-up clinical values. For each baseline snapshot, a configuration and seed define time offsets, supported treatment exposure metadata, tumor-burden curve and independently varying symptoms, lab trends and adverse-effect indicators. Preserve all source baseline values, including missingness.

Five supported **hypothetical** families from COS-SPEC-004:

1. Sustained tumor shrinkage.
2. Limited improvement / relative stability.
3. Increasing tumor burden.
4. Initial improvement followed by regrowth.
5. Deterioration / treatment-related toxicity pattern that need not correlate with tumor burden.

Mandatory checks: monotonic time order, no negative physical volumes, nonnegative counts where appropriate, valid measurement units, clinically constrained treatments only when

qualifying source status is available, no independent-patient inflation, compatible mesh volume within a defined geometric tolerance, seed reproducibility, no authentic-data overwrite, explicit branch label, and publication only after schema and provenance validation.

Split leakage: baseline patient and every derivative synthetic branch must reside in the same model train/validation/test split. The same applies to derivative datasets from overlapping TCIA cohorts. Synthetic test branches cannot demonstrate generalization to unrelated real patient outcomes.

10. AI interpretation, RAG and external inference

Separation of concerns: imaging code generates validated lesion measurements; tabular code computes time differences, deltas, appropriately contextual reference-range comparisons, and missing-data warnings; NLP extracts explicit clinical facts; retrieval produces cited evidence snippets; the language model narrates *only* the verified factual payload.

Seven logical components: image intelligence; note extraction; tabular analysis; temporal state aggregation; evidence retrieval; multimodal fusion; interpretation generation. They can live in separate application packages and worker tasks, not seven network services.

LLM execution contract:

- ai_gateway sends permitted, de-identified, minimized state summaries and retrieved evidence through an adapter such as MetaModelApiAdapter; configure provider, model_id, and request limits outside code.
- Prompt injection defenses: treat notes, retrieved passages and external model output as data; disallow privileged tool calls and unverified instructions embedded in those sources.
- Require structured schema with finding, value_reference_ids, evidence_reference_ids, source_type, uncertainty, limitations, model_id, created_at. Reject unresolved citations or unsupported numerical values.
- Never let language-model responses mutate baseline clinical tables, launch treatments, issue clinical recommendations or request arbitrary SQL.
- Do not transmit raw DICOM or possibly identifying clinical notes externally without explicit, documented authorization and vendor data-processing review.
- Cache only inputs/outputs that policy permits, by patient state + source hash + prompt/model/version, with invalidation when inputs change.
- Provide provider timeout, exponential backoff, fixed retry budget, spend cap and a deterministic no-LLM fallback summary.

RAG: start with an approved, small curated literature collection (version, date, URL/DOI, sections, license). PostgreSQL full-text search is enough for first delivery; add pgvector embeddings only if retrieval evaluation shows benefit. Evidence passages explain relevant scientific context, **not** a personalized prescribing conclusion. Link every citation to a document and exact passage metadata.

Meta Model API verification: official Meta documentation currently lists Muse Spark models and structured-output capabilities; model versions, account access, quota, privacy mode and

current prices must be checked at deployment. Do not assume a generic SAM image model performs clinically validated breast MRI segmentation.

11. REST API specification and permissions

Prefix /api/v1; expose OpenAPI definitions and versioned JSON schemas. GET operations are side-effect free; mutating requests use authentication, role guards, content validation, body-size limits and audit logging. Job execution is asynchronous and idempotent by key where appropriate.

Method and route	Actor	Contract / outcome
GET /cases	viewer+	Page approved pseudonymous case summaries
GET /cases/{id}	viewer+	Case diagnosis summary, missingness and provenance
GET /cases/{id}/states	viewer+	Observed / simulated states with separate branch IDs
GET /states/{id}	viewer+	Full typed patient-state manifest with reference IDs
GET /states/{id}/observations	viewer+	Date-sorted labs, vitals, symptoms, treatment events
GET /states/{id}/geometry	viewer+	Mesh metadata and authorized short-lived download URL
GET /studies/{id}/series	viewer+	MRI acquisition metadata (not direct sensitive pixel exposure)
GET /studies/{id}/measurements	viewer+	QC'd physical measurements and annotation class
POST /imports	admin	Signed upload/import job from an approved source manifest
POST /studies/{id}/process	admin/editor as allowed	Queue validated MRI preparation job
POST /studies/{id}/segment	admin only in P1	Optional evaluation-gated segmentation job
POST /simulations	editor+	Start configured,

Method and route	Actor	Contract / outcome
		reproducible branch generation
GET /simulations/{id}	viewer+	Branch outputs, config, source baseline, validation
POST /cases/{id}/observations	editor+	Append new research observation with attribution; no original edit
POST /interpretations	editor+	Queue evidence-linked research narrative
GET /interpretations/{id}	viewer+	Validated result, references, provenance, limitations
GET /jobs/{id}	permitted requester/admin	Status, stage, error category and retry information
GET /evidence/{id}	viewer+	Allowed citation metadata and linked passages
GET /admin/validation-results	admin	QA findings and release gate

Standard async response: HTTP 202 Accepted with {job_id, status, status_url, correlation_id}. **Standard error:** Problem Details style {type,title,status,detail,instance,correlation_id}; do not leak paths, tokens or patient secrets. For concurrent publication use transactional version checks and unique idempotency tokens. Query-level case authorization is mandatory, not merely role-level authentication.

11.1 Example typed patient-state envelope


```
{
  "schema_version": "1.0",
  "patient_id": "uuid",
  "state_id": "uuid",
  "state_type": "SYNTHETIC",
  "observation_time": "2030-04-01T00:00:00Z",
  "branch_id": "uuid",
  "source_baseline_state_id": "uuid",
  "modalities": {"imaging": "DERIVED", "laboratories": "SYNTHETIC", "notes":
"UNAVAILABLE"},
  "measurements": [{"concept": "lesion_volume", "value": 8.1, "unit": "cm3",
```

"provenance_id": "uuid"}],
  "geometry_artifact_id": "uuid",
  "validation_status": "PASSED",
  "limitations": ["Hypothetical branch: not an observed patient outcome"]
}

The timestamp and lesion volume above are **purely illustrative contract values**, not patient findings.

12. Job orchestration, reliability and performance

Worker classes: ingestion-worker, imaging-worker (CPU baseline), simulation-worker, interpretation-worker; keep separate resource limits even when implemented by the same source code. One Redis-backed queue is sufficient initially. For reliability, persist authoritative job status and results in PostgreSQL; Redis is operational transport/cache and may be recreated. Use worker acknowledgments after persistence, timeout limits, retryable/nonretryable error taxonomy, deduplication and a dead-letter/failed-job queue.

Idempotency keys: (source_artifact_sha256, pipeline_version, config_hash) for imaging; (baseline_state_id, scenario, parameter_hash, seed, simulator_version) for simulation; (state_hash, evidence_set_hash, prompt_version, model_id) for interpretation.

Performance SLOs (engineering targets; not measured facts): normal metadata reads p95 under 700 ms once warmed; authorized state manifest p95 under 1.5 s; first usable mesh view under 5 s for a precomputed demo-sized artifact on an adequate connection; all lengthy computation returns a job identifier promptly. Record cold-start and mobile performance separately. Do not state GPU segmentation speed until benchmarked with chosen volumes/models.

Failure behavior: API and data browsing must remain available during LLM provider outage. Geometry can remain available if simulation queue is down. If image preprocessing fails, source remains quarantined and no new clinical volume is published. Prior validated states are retained across all failures.

13. Security, privacy, data rights and clinical safety

- Require HTTPS, secure auth/session management, CSRF protection where cookies are used, minimum-privilege roles and object access scoped to the authenticated case.
- Secrets in managed secret storage or protected deployment environment; no tokens in browser bundles, logs or repository.
- Encrypt database and object storage at rest; restrict administrator access; log all sensitive reads and mutation attempts.
- Disable unreviewed public uploads; restrict file types, file size, archive expansion, DICOM metadata and path traversal; scan uploads/quarantine where practical.
- Maintain a dataset-license register with proof of source, permitted uses, attribution, redistribution limits and commercialization review. CC BY-NC assets must not silently flow into an unrestricted commercial deployment.

- Pseudonymize source identifiers and establish small-sample privacy review: de-identification does not always eliminate re-identification risk, especially where images and rare attributes are linked.
- Treat medical software compliance as a future legal/regulatory question dependent on intended use and jurisdiction; do not claim the prototype is approved for clinical care.
- Ensure public demo displays unmistakable research-only disclosure, source labels, AI limitations and no treatment-selection interface.
- Maintain backups with retention and restoration checks, access/event logs, dependency scanning and optional SBOM for submitted artifacts.

14. Local development environment

Known constraint: AMD Ryzen 5 3500U-class laptop, 8 GB RAM, SSD and low-memory integrated/discrete graphics in the developer's current setup. GPU vendor/model memory details and real CUDA capability must be benchmarked rather than assumed. Plan for CPU-first development.

Local minimum workflow: VS Code + Git, Python virtual environment, Node/Next.js and Docker only for PostgreSQL/Redis if system memory permits. Develop FastAPI and Next.js natively (or one container at a time), seed with a few compressed NIfTI/mask/mesh demo assets, and keep large medical-image archives and checkpoints off the laptop. Precompute mesh assets on a remote CPU worker if necessary. Do not run a multi-service Kubernetes cluster, a large LLM, 3D MRI training and a modern browser simultaneously on 8 GB RAM.

Recommended improvement: 16 GB RAM materially improves local API/frontend/browser/container concurrency; 32 GB is useful for full-volume imaging workflows. An NVIDIA GPU with suitable VRAM and CUDA stack is only justified when pursuing automated MRI segmentation training/inference; 2 GB VRAM is not a credible default assumption for full 3D DCE-MRI neural inference.

Developer test suite: tiny fake research cases and synthetic MRI arrays generated exclusively for engineering tests, plus separately licensed verified real demo samples. No invented medical facts in the public research demo.

15. Cloud deployment topology

15.1 Preferred hackathon topology — low-operations cloud

- Frontend: static/SSR-capable Next.js hosting with TLS and environment-based API origin.
- API: one CPU container deployment for FastAPI, 1–2 initial replicas as affordable.
- Queue workers: one on-demand or small CPU worker deployment; separate heavier MRI jobs by resource class.
- PostgreSQL: managed entry-tier private database, with automated backup and migration process.
- Redis: small managed instance or co-located single-node development queue only for an isolated prototype.

- Object storage: S3-compatible private bucket with signed access; choose a provider/region consistent with policy and deployment requirements.
- AI: outbound HTTPS through backend-only hosted Model API adapter, with usage and privacy gate.
- Optional segmentation: on-demand GPU execution **only when** pretrained checkpoint and held-out evaluation justify it.
- Monitoring: error monitoring, structured event logs, job metrics and a hard cloud budget alert.

Azure reference: Azure Container Apps Consumption can host API/CPU worker containers and jobs, and may scale to zero. Microsoft documents consumption-based billing and an included free monthly usage allowance for selected CPU/HTTP meters; this is **not** a statement that a whole CancerOS stack will be free. Managed Postgres, storage, network and GPU can create charges. Region-specific GPU availability and quota must be checked independently.

Alternative portability: deploy the same OCI containers on any suitable VM/container host; retain PostgreSQL and S3-compatible object APIs. Avoid Azure-only logic in clinical modules. Do not introduce Kubernetes, service mesh or specialized medical imaging server unless a later phase demonstrates a need.

15.2 Approximate starting resource envelope (capacity assumptions; not benchmarks)

Component	Trial sizing assumption	Validation requirement
Next.js	Managed hosting; small SSR instance when needed	Real-world route, hydration and JS memory benchmark
FastAPI	1–2 vCPU, 1–2 GB RAM	Load test case endpoints and schemas
Postgres	Small managed instance, 1–2 GB+ RAM, modest SSD	Connection and query throughput; backup success
Queue/Redis	Small managed tier	Retry and persistence under restarts
Imaging CPU worker	2–4 vCPU and 8–16 GB RAM **initial benchmark candidate**	Peak resident memory on representative DCE-MRI volumes
Optional GPU job	No always-on GPU; on-demand selected hardware	Checkpoint VRAM test, cost / completion
Object storage	Measured sample size + processing copies + backups	Actual bytes per selected case; egress policy
External LLM	Per-request billing and rate limits	Prompt lengths, approved payload, schema success and failure rate

The imaging worker's peak RAM could exceed the provisional 8–16 GB assumption depending on source volume shape, number of DCE phases and preprocessing copies. Establish an empirical upper bound before accepting it as final sizing.

15.3 Budget calculation method

Do **not** publish a flat monthly dollar amount without specific provider, region, instance type, concurrency, storage and inference usage. Monthly cost is approximately:

frontend + API compute + worker CPU-seconds + DB uptime/storage + Redis + GB-months stored + operation counts + transfer/egress + LLM input/output tokens + optional GPU seconds + backups/monitoring.

For early research demos, precomputing five curated patient twins often saves more money than provisioning a continuously running GPU. Record actual resource usage and request counts for a representative public-demo session, then multiply by anticipated sessions and include a safety margin. Configure spend alerts and a manual disable switch for inference jobs.

16. Build, release and CI/CD

Repository structure:

```
canceros/
  apps/web/                 # Next.js UI
  services/api/             # FastAPI entrypoint, auth, REST
  packages/contracts/       # OpenAPI, JSON schemas, generated TS types
  packages/clinical/        # canonical domain, provenance and temporal logic
  packages/imaging/         # MRI/mask QC, physical measurements, mesh
  packages/simulation/      # seeded longitudinal engine
  packages/intelligence/    # extraction, evidence and interpretation
  workers/                  # queue execution entrypoints
  infrastructure/           # docker, IaC, deploy manifests
  migrations/               # DB migrations
  tests/                    # unit, integration, scientific regression
  fixtures/                 # synthetic code-test only; never real PHI
  docs/                     # data registry, licenses, runbooks
```

CI gates: format/lint/type checks; unit tests; migrations against temporary DB; API OpenAPI snapshot compatibility; scientific volume/affine regression tests; simulation seed reproducibility; source/synthetic separation tests; AI contract tests with mocked provider; object permission tests; dependency and secrets scans. Package pinned versions and immutable image tags. Promote a staging deployment before publicly exposing the demo. Roll back images and DB-compatible changes if needed.

Release artifacts: Dockerfiles, compose development stack, deployment instructions, data-source/license registry, configuration template, no-secret demo fixture, processing manifests, reproducibility report, research-only disclaimer and screenshots/video backup of demonstration.

17. End-to-end acceptance plan

P0 — must work without specialist GPU

1. Import one eligible real breast MRI with verified source rights and mask (or appropriate documented annotation), preserving original files and provenance.
2. Prove physical-space geometry correctness with unit tests and independent overlay inspection.
3. Display rotatable lesion mesh and correct measurement in a browser.
4. Produce five distinct labeled synthetic branches with at least four dated states each, using one seed/config and reproducible outputs.
5. Show clinical context with explicit missingness; demonstrate that labs/notes that do not exist in source are not presented as observed.
6. Compare two dates numerically with traceable source/derived references.
7. Generate a structured research interpretation referencing real state IDs and curated evidence; if LLM is offline, return a deterministic honest summary.
8. Demonstrate one **genuine** repeated-imaging case where linked visit dates, masks and comparable measurements were actually validated; if source matching fails, flag unmet requirement rather than substituting simulated dates.
9. Enforce viewer/editor/admin permissions and short-lived asset URLs.
10. Redeploy from committed source and reproduce a complete demo from documented steps.

P1 — only after P0 is solid

Automated breast MRI segmentation with held-out evaluation; interactive mask correction; richer 2D/3D linked imaging; registration enhancements; broader real-case cohort; advanced RAG; optional learned temporal model trained/evaluated using patient-independent real data.

Explicitly deferred

Whole-body physiology; patient-specific treatment outcome estimation as validated fact; radiotherapy plans; autonomous diagnostic or prescribing workflows; multi-cancer expansion; hospital EHR integration; standalone graph database; distributed Kubernetes architecture.

18. Phased implementation roadmap

Phase	Output	Hard exit gate
0 — Data and model due diligence	Verified license/source manifests, selected sample, mask semantics, LLM access decision	At least one legally usable geometrically compatible example
1 — Vertical slice	FastAPI + PostgreSQL + Next.js + private asset delivery + patient case view	Real MRI-derived lesion successfully rendered
2 — Longitudinal and	State/branch schemas,	Repeatable result and

Phase	Output	Hard exit gate
simulation	five scenario rules, deterministic variation and provenance	zero observed/synthetic confusion
3 — Multimodal intelligence	Lab/vital changes, evidence retrieval, approved AI gateway and narrative guardrails	Every supported claim/source ID traceable, invalid claims rejected
4 — Authentic temporal proof	Linked repeated-MRI case, comparable measurements, discrepancy report	Two or more valid authentic timepoints displayed with verifiable sources
5 — Hardening and deployment	Staging/production scripts, auth, privacy, metrics and demo rehearsal	Fresh deploy, failure fallback and release checklist pass

First vertical slice should not depend on 100 cases, full dataset download, nnU-Net training, real-time agent orchestration, a graph database or a paid GPU reservation.

19. Risks and mitigations

Risk	Severity	Mitigation and evidence gate
No verified mask or misidentified analysis mask	Critical	Source annotation inspection; reference-mask-first P0; no volume claim without valid boundary
Cross-cohort duplication / training leakage	Critical	Preserve original TCIA identifiers; same parent-group split; dedup registry
Dates or DCE phases misrepresented as real follow-up	Critical	Imaging acquisition vs clinical visit schema; temporal checks
Synthetic output mistaken for prediction	Critical	Type tags in DB, API and UI; persistent research limitation text
Dataset commercialization conflict	High	License-aware asset registry and deployment review, especially CC BY-NC sources
LLM invents measurements or	High	Deterministic measurements; allowed-

Risk	Severity	Mitigation and evidence gate
citation		reference validator; mock/error fallback
External AI receives sensitive records	High	Deny-by-default export policy; minimal payload; provider/legal review
8 GB local laptop OOM	High	Small samples; remote preprocessing; 16 GB upgrade if affordable; limit concurrent services
GPU costs/cold start prevent demo	High	Precompute P0 masks/meshes; optional GPU path only
Cloud bills exceed budget	Medium-high	Metered budgets, capped queues, scale-to-zero where appropriate
Poor rendering on common devices	Medium	GLB LOD, mesh size limits, graceful fallback and mobile testing
Overengineering delays integrated result	High	Modular monolith, P0 first, explicitly deferred services

20. Architecture decision records

ADR-001 — Python FastAPI over C# as primary scientific backend. Chosen to minimize inter-language barriers for SimpleITK, NiBabel, NumPy, PyTorch/MONAI and scientific computation. C#/.NET remains technically capable for enterprise APIs and could be introduced for separate operational needs, but a dual-language distributed backend is not justified during the hackathon.

ADR-002 — PostgreSQL rather than a graph database for P0. Knowledge associations are initially source/evidence and patient-state relationships with well-defined schema; use relational edge tables and optional vector extension. Introduce a dedicated graph store only for proven query workloads.

ADR-003 — Precomputed web meshes rather than browser MRI segmentation. Dramatically reduces browser compute, download volume and source image exposure while keeping scientifically measured geometry on the backend.

ADR-004 — Separate AI narrative from measurement computation. Numeric reliability and source traceability are validated in deterministic code; LLM is constrained to extraction and explanations.

ADR-005 — No mandatory GPU in minimum deploy. Reference masks, CPU image processing, synthetic simulation and external hosted LLM are sufficient for the core demo. Optional medical segmentation uses batch GPU if independently evaluated.

ADR-006 — Immutable observed record and append-only synthetic descendants. Essential to prevent simulation output from contaminating real-data interpretation and future evaluation.

ADR-007 — Provider-neutral storage and LLM adapters. Reduces switching costs if current cloud grants, Meta API access or privacy requirements change.

21. Unresolved items requiring verification before final implementation

- Which exact I-SPY2/I-SPY1 patient studies have complete matched dates, suitable annotations, allowed access and comparable physical measurements?
- Are chosen imported masks expert lesion contours, model-derived anatomical contours, bounding annotations, or functional-tumor-volume analysis regions?
- Is use of selected CC BY-NC assets permitted within the exact hackathon rules, public demonstration and future commercialization scenarios?
- Which clinical variables really exist for selected source cases (particularly labs, vitals, symptoms and clinician notes), and which must be clearly synthetic?
- What is current Meta API account eligibility, chosen model identifier, rate limit, price, logging/retention, allowed health-data handling and reliability against a held-out medical-note test set?
- What worker RAM and CPU time do representative MRI files and selected mesh extraction parameters actually require?
- Which cloud region/provider offers authorized data handling, affordable managed Postgres and convenient worker processing for the specific account?
- Does a useful pretrained breast DCE-MRI segmentation checkpoint actually exist under an appropriate license, with demonstrated transfer to the selected source distribution?
- What actual privacy, clinical safety and hackathon organizer obligations apply to the intended hosted public demo?

These are **open decisions**, not facts that should be guessed into procurement or software architecture.

22. References and traceability

Supplied requirements (proposed, not independently validated by their existence)

- **COS-PRD-001** — *CancerOS Product Requirements Document*, 22 October 2026.
- **COS-SPEC-003** — *Clinical Data & Dataset Specification*, 22 October 2026.
- **COS-SPEC-004** — *Synthetic Longitudinal Data Specification*, 22 October 2026.
- **COS-SPEC-005** — *Medical Imaging & Digital Twin Specification*, 22 October 2026.
- **COS-SPEC-006** — *AI & Multimodal Intelligence Specification*, 22 October 2026.

External checked primary references (accessed 22 October 2026)

1. TCIA, I-SPY2 collection: https://www.cancerimagingarchive.net/collection/ispy2/

2. TCIA, Duke Breast Cancer MRI: https://www.cancerimagingarchive.net/collection/duke-breast-cancer-mri/
3. Zenodo, BreastDCEDL release and rights: https://zenodo.org/records/17274053
4. Meta Model API overview: https://dev.meta.ai/docs/overview
5. Meta Model API model catalog: https://dev.meta.ai/docs/models
6. OHIF official documentation: https://docs.ohif.org/
7. Microsoft, Azure Container Apps billing: https://learn.microsoft.com/en-us/azure/container-apps/billing
8. Microsoft, Azure Container Apps workload profiles: https://learn.microsoft.com/en-us/azure/container-apps/workload-profiles-overview
9. Cloudflare R2 official pricing and limits: https://developers.cloudflare.com/r2/pricing/

Evidence posture: Official provider pages support the claims noted in the verification ledger. Specific cloud bills, model accuracy, dataset subset completeness, medical validity and the project's implementation outcomes remain unverified until measured or evaluated.